

Geometry-Centered Quantum Foundations and Structural Quantum Mechanics

Structural Quantum Mechanics (SQM), as you drafted it on April 17, 2025, is best read as a **geometry-first, wavefunction-realist interpretation**: the wavefunction is taken to be a real relational structure, and “measurement/decoherence/classicality” are recast as **stabilizations and reconfigurations of relational geometry** rather than literal collapse events. When evaluated against the existing foundations and quantum-information literature, SQM sits in a well-populated (and intellectually serious) region: **geometric quantum mechanics, open quantum systems/decoherence, ψ -ontology debates, and relational/constraint-based** reformulations of what quantum theory is “about.” ¹

Geometry as the primary language of quantum theory

A large body of work formalizes what you’re intuitively aiming at: **quantum theory as geometry on the manifold of states**. In geometric quantum mechanics, the physically distinct pure states are not vectors but **rays** (global phase is unphysical), so the “space of pure states” is naturally modeled as a **complex projective space** equipped with a unitary-invariant metric—the **Fubini-Study metric**—and an associated symplectic structure. In this setting, the Schrödinger equation can be rewritten as **Hamiltonian flow on projective Hilbert space**; observables become geometric objects (functions generating flows) rather than “things in space.” ²

This geometric formulation matters to SQM because it supports two central SQM intuitions in a technically orthodox way:

- “Structure rather than substance”: the physically relevant content is in **relations among amplitudes** (inner products, phases, interference structure), which are explicitly geometric on the state manifold. ³
- “Dynamics as motion through a constrained space”: unitary evolution preserves the projective geometry (it’s an isometry for the Fubini-Study structure), so “laws” can be read as **constraint-preserving flows** on a curved state manifold. ⁴

A key bridge from “geometry talk” to operational physics is that the same geometric ideas reappear as **distinguishability geometry**: metrics on state space that quantify how well two states can be told apart by experiments. That thread becomes crucial once you move from pure-state/unitary geometry to **mixed states and noise**, i.e., decoherence. ⁵

Decoherence and open-system evolution as geometric deformation

SQM’s strongest contact point with mainstream physics is its framing of decoherence as **reconfiguration rather than collapse**. In standard open-systems theory, a system interacting with an environment is described (in general) by a **completely positive trace-preserving (CPTP) map** (a quantum channel), which can be written in a **Kraus operator-sum representation**. ⁶

The geometric core of the “no collapse needed” view is captured cleanly by the **Stinespring dilation** idea (sometimes summarized as the “larger Hilbert space” viewpoint): any physically allowed open-system evolution can be represented as **unitary evolution on system+environment** followed by “forgetting” (tracing out) the environment. In that sense, what looks like non-unitary “loss of coherence” for the subsystem is consistent with globally unitary evolution—exactly the intuition you express when you say the state “changes form” rather than collapses. ⁷

Within this standard framework, “decoherence follows pathways” becomes concrete via **environment-induced superselection (einselection)** and the emergence of **pointer states**: the environment effectively monitors certain observables, suppressing interference between preferred states and making those states stable under environmental coupling. This is not a vague metaphor in the literature; it’s developed as a mechanism that restricts which sectors of Hilbert space remain robust under realistic interactions. ⁸

Your SQM language—“filtering,” “constraint-selective,” “stabilized relational subspace”—maps closely onto this: the environment selects (by dynamical stability) a narrow family of robust correlations, and classical-looking records are the stable ones. ⁹

From a simulation standpoint, the thing you report observing (different decoherence “shapes” under amplitude damping vs phase damping vs depolarizing noise) is a baked-in consequence of the **different Kraus structures** of those channels. This is exactly why gate-model noise modeling is expressed in Kraus or channel form in both theory and tooling. ¹⁰

Your use of Qiskit for this is well aligned with how the field operationalizes such questions: Qiskit Aer provides explicit machinery to construct noise models and channels for simulation, including standard damping and depolarizing-type errors. ¹¹

Relational realism and the wavefunction’s ontic status

SQM takes a clear stand: the universal wavefunction is **ontic** (real), while “particles/measurements” are emergent patterns of relational structure. In the foundations literature, this stance is not merely philosophical; it intersects with formal results known as **ψ -ontology theorems**, which (under explicit assumptions) constrain interpretations where the quantum state is “mere information.” The most famous of these is the PBR result, frequently summarized as showing that, given preparation-independence assumptions, the wavefunction cannot be treated as purely epistemic in the relevant hidden-variable sense. ¹²

SQM’s “relational encoding” also resonates with **relational quantum mechanics (RQM)**, which rejects an observer-independent state while avoiding the requirement that “observer” means “conscious mind.” RQM instead treats quantum states as descriptions of relations between physical systems. This is philosophically close to SQM’s “structure is fundamental,” but note the subtle difference: RQM is often read as downgrading a single universal state description in favor of relational state assignments, whereas SQM (as written) emphasizes a **universal wavefunction as global structure**. ¹³

It is also a live topic how ψ -ontology arguments interact with relational frameworks; there are published analyses arguing that certain relational views are not straightforwardly targeted by PBR-style assumptions because the assumptions themselves presuppose a particular notion of preparation/state attribution. ¹⁴

In short, SQM's wavefunction realism is strongly "in-family" with a significant slice of modern foundations discourse, and its relational emphasis has clear neighbors (RQM, information-theoretic reconstructions), but SQM's distinctive flavor is the insistence that the *primary* explanatory object is **geometry of constraints** on the state space. ¹⁵

Metrics, distinguishability, and why noise looks like geometric filtering

A powerful way to make "constraint-selective geometry" mathematically sharp is to lean into **state-space metrics** and their behavior under physical evolution.

One landmark result in this direction is the program that defines the geometry of quantum states via **statistical distinguishability**: by considering optimal measurements for telling nearby states apart, one gets a natural Riemannian metric on the space of density operators (mixed states included). ¹⁶

Closely related is the **Bures metric** (also called the Helstrom metric in some contexts), which is widely used in quantum information geometry and agrees with the Fubini–Study metric on the pure-state submanifold. ¹⁷

Here is the key connection to SQM's decoherence language:

Physical open-system evolution is modeled by CPTP maps (channels), and many operationally meaningful distinguishability measures are **contractive under CPTP maps**—they can only stay the same or shrink. That contraction is a rigorous sense in which noise "compresses" the effective geometry: it reduces distinguishability, suppresses off-diagonal coherence in suitable representations, and drives large families of initial states toward smaller families of effective states. ¹⁸

This contractivity appears both in common distances like trace distance (often used in non-Markovianity diagnostics) and in families of monotone Riemannian metrics studied in mathematical physics and quantum information. ¹⁹

From an SQM perspective, you can reinterpret these standard facts as:

- the environment does not "randomly destroy structure"; it applies a **physically admissible contraction** (a constrained map) on the state manifold,
- and classicality corresponds to **stable attractors/fixed sets** (pointer structures) of those contractions. ²⁰

This is also where your phrase "topology shift" can be made precise. Many qualitative "topology-like" changes in open dynamics correspond to changes in rank, support, or connectivity of distinguishability structure (e.g., whether interference terms are effectively accessible), and information-geometry work explicitly studies curvature and geometric features of quantum state space under physically meaningful metrics. ²¹

Emergent time as a constraint- and state-dependent notion

Your SQM section “Time as Emergent” is ambitious, but it is not unmoored from existing technical proposals. Two major research traditions are especially relevant.

One is the **thermal time hypothesis**, associated with the algebraic approach to quantum theory: time flow is treated as **state-dependent**, derived from structural properties of von Neumann algebras (via Tomita–Takesaki theory), rather than assumed as a background parameter. This explicitly frames time as emerging from the state/constraints of the system. ²²

Another is the **Page–Wootters mechanism**, where time is argued to emerge relationally from entanglement/correlations between a subsystem (“clock”) and the rest. There are experimental illustrations of this mechanism and modern theoretical developments applying it to recover classical equations of motion in suitable limits. ²³

These programs are not “the same as SQM,” but they show that “time as emergent from relational structure” is a technically developed theme rather than a purely philosophical gesture. They also naturally fit your **geometry-first** instincts because both approaches emphasize relational/state-dependent structure over background primitives. ²⁴

SQM’s closest research neighbors and what would make it scientifically distinctive

When SQM is placed against the literature above, the most accurate taxonomy is:

SQM (as written) is primarily an **interpretation** that (i) is wavefunction-realist, (ii) re-describes decoherence in geometric/structural terms consistent with dilation/open-system theory, and (iii) treats classicality as an emergent stability phenomenon (pointer-like subspaces). ²⁵

That is already valuable if your goal is “geometry itself,” because it pushes you toward the mature mathematical objects where geometry lives: projective Hilbert space, Fubini–Study/Bures families of metrics, CPTP maps, and stability/attractor structure of Lindblad generators. ²⁶

The hard part (and the opportunity) is your SQM claim of **empirical leverage**: “structurally determined pathways” is true in standard decoherence theory, so SQM becomes scientifically distinctive only if it contributes at least one of the following (each has direct anchors in the literature):

- **A precise geometric invariant** (or family of invariants) that predicts classes of decoherence trajectories across noise models or system symmetries, beyond simply re-stating the Kraus/Lindblad calculation. Work that pulls back the Fubini–Study metric to local-unitary orbits and uses it to build entanglement-related invariants is an existence proof that “geometric invariants for quantum information structure” can be made algorithmic. ²⁷
- **A principled geometric meaning of “constraint network”** in terms of channel structure: for example, reading decoherence as contraction in a chosen monotone metric (Bures/trace/etc.), and characterizing “classical subspaces” as stable sets/fixed points of the channel dynamics. The

contraction/monotonicity results for distances and metrics under CPTP maps provide the technical backbone for exactly this “filtering” language. ²⁸

- **A bridge to topological/robust encodings** where “geometry over substance” becomes operationally unmistakable. In topological quantum computation, information is stored nonlocally in topological degrees of freedom and manipulated by braiding, giving a concrete sense in which *global structure* matters more than local coordinates—strongly aligned with SQM’s rhetoric even if the underlying physics is specialized condensed matter. ²⁹

Finally, your SQM nod to constructor-theoretic language (“possible vs impossible transformations”) has a clear neighbor in the constructor theory literature, which explicitly reframes fundamentals in terms of constraints on tasks rather than trajectories. If SQM is “constraints evolving in the wavefunction,” constructor theory is a broad attempt to make “constraint statements” the primitive explanatory currency.

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